| COURSE INFORMATION SHEET | | | |
|---|---|---|---|
| Program | **ELECTRONICS ENGINEERING** | | |
| Course | **Embedded Systems** | | |
| Course Code | **5041** | Semester | **5** |
| Revision | **2021** | Credits | **4** |
| Course Type | **THEORY** | Course Category | **Program Core** |
| Periods per week | **4** | Periods Per semester | **60** |
| Corresponding Lab course Code (If any): | **5147** | Lab course Name | **Embedded Systems Lab** |

**Syllabus:**

| Module | Description | Duration (Hours) |
|---|---|---|
| I | **Embedded Systems** - Definition, difference from general purpose computers -Classification of embedded systems, Application areas,<br>**Components of embedded system**hardware, andSoftware embedded into the system<br>**Architecture of embedded system** – Building blocks of an embedded system , Core of embedded system – categories, Memory –ROM and RAM, Sensors, actuators and I/O sub systems (LED, Opto coupler, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wifi) | 12 |
| II | **AVR Microcontroller Architecture** - Comparison of AVR family members and Selection of a microcontroller, ATMega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory -SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers-0,1,2 (Block level) associated registers, Interrupts.<br>**AVR programming using embedded C:** Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2, AVR interrupts - programming of timer interrupts - programming external hardware interrupts - interrupt priority in AVR microcontroller. | 19 |
| III | **Interfacing** of LED, Push button, Relay, Opto Coupler, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), RTC Interfacing, ADC interfacing<br>On-board communication interfaces with AVR – I2C interfacing(like real time clock interfacing)and basics of SPI interfacing with AVR. | 12 |

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|---|---|---|
| IV | **Operating system basics**:- Kernel, types of operating systems- GPOS, RTOS. **Real time operating systems**:- Tasks, process, threads, multiprocessing and multi-tasking, task scheduling, types, threads and process scheduling, task communication, task synchronization, device drivers.List Selection criteria for RTOS. Overview of Micro C/OS-II and its services.<br>List popular Real Time Operating Systems (any 10). | 15 |
| | **TOTAL HOURS** | **60** |

**Text / Reference books:**

| T / R | BOOK TITLE/ AUTHORS/ PUBLICATION |
|---|---|
| T1 | 1. The AVR Microcontroller and Embedded Systems Using Assembly and C By Muhammad Ali Mazidi, Sarmad Naimi, & Sepehr Naimi - Pearson Education |
| R1 | Introduction to Embedded Systems - Shibu K.V, Mc Graw Hill.First Edition<br>Embedded Systems - Raj Kamal, , Mc Graw Hill, Second Edition |
| R2 | Embedded C - Michael J. Pont, Pearson Education, Second Edition |
| R3 | Embedded Systems - Raj Kamal, , Mc Graw Hill, Second Edition |

**Course Prerequisites:**

| Course Code / Topic | Course Name | Description | Semester |
|---|---|---|---|
| programming concepts | Programming in C - 3045 | Concept of C programming | III |
| Programming Concepts | Python Programming – 4049 | Comcep[t of Python programming | IV |
| Basics of Microcontroller architecture | Microcontroller and Application - 4041 | Architecture and concept of MC | IV |

**Course Objectives:**

| | |
|---|---|
| I | Introduce the importance of embedded systems |
| II | Introduce ATMEGA 32 microcontroller and its suitability for embedded systems |
| III | Familiarize embedded C programming |
| IV | Familiarize serial communication and data transfer between embedded systems |

**Course Outcomes:**

| No. | Description | Cognitive Level |
|---|---|---|
| I | Explain the basics of embedded systems and its architecture | Understanding |
| II | Make use of AVR Microcontrollers to develop embedded programs using embedded C | Applying |

| | | |
|---|---|---|
| III | Make use of AVR microcontroller to interface with various peripheral devices. | Applying |
| IV | Familiarize RTOS | Understanding |

**CO-PO & CO-PSO Mapping:**

| | PO 1 | PO 2 | PO 3 | PO 4 | PO 5 | PO 6 | PO 7 | PSO 1 | PSO 2 | PSO 3 |
|---|---|---|---|---|---|---|---|---|---|---|
| CO 1 | **2** | | | | | | | 2 | | |
| CO 2 | **3** | 3 | | | | | | | | |
| CO 3 | **3** | 3 | | | | | | | | 3 |
| CO 4 | **3** | | | | | | | | | |

**Justification for CO-PO Mapping:**

| Mapping | Level | Justification |
|---|---|---|
| CO1-PO1 | 2 | Students should apply the knowledge of mathematics, science, Engineering fundamentals, and Electronics / computer Engineering |
| CO2-PO1 | 3 | Problem analysis is required in the design of interfacing circuits and systems using microcontrollers. |
| CO3-PO1 | 3 | The programming of microcontrollers is always started with the problem analysis and related with interfacing |
| CO4-PO2 | 2 | Embedded systems follows RTOS concepts in UI & resource management |

**Gaps in the syllabus to meet Industrial / Profesional Requirements :**

| No. | Description | Relevance to PO & PSO | Actions Proposed |
|---|---|---|---|
| 1 | Embedded systems professional design and development methodologies are not included in the current syllabus | PO7, PO6 | Workshops and seminars<br>Mini Projects |
| 2 | Familiarization of modern design tools and platforms | PO7,PO6 | Workshops and Industrial visits |

**Topics beyond Syllabus & Advanced Topics :**

| No. | DESCRIPTION | Relevance to PO & PSO | Proposed Actions |
|---|---|---|---|
| 1 | Modern design and simulation tools | PO6, PO7 | Workshops |

| | | | |
|---|---|---|---|
| 2 | Embedded systems and IOt | PO7 | seminars |

**Web / Resources / References :**

| | |
|---|---|
| 1 | www.nptel.com |
| 2 | www.atmel.com |
| 3 | www.intel.com |
| 4 | www.microchip.com |
| 5 | www.arduino.org |
| 6 | www.raspberripi.org |

**Delivery / Instructional Methods :**

| | | |
|---|---|---|
| ■ CHALK &TALK | ■ STUDENT ASSIGNMENT | ■ WEB RESOURCES |
| ■ LCD/ SMART BOARD | ■ STUDENT SEMINARS | □ ADD-ON COURSES |

**Assessment Methodologies ( direct):**

| | | |
|---|---|---|
| ■SIGNMENTS | □ STUDENT SEMINARS | ■EST / MODEL EXAMS |
| ■ BOARD EXAMINATION | ■ STUDENT LAB PRACTICES | ■TUDENT VIVA |
| □ MINI / MAJOR PROJECTS | □ CERTIFICATION | □ OTHERS |

**Assessment Methodologies ( Indirect):**

| | |
|---|---|
| ■ ASSESSMENT OF COURSE OUTCOMES (BY FEEDBACK, ONCE) | ■ STUDENT FEEDBACK ON FACULTY (ONCE) |
| □ ASSESSMENT OF MINI/ MAJOR PROJECTS BY EXTERNAL EXPERTS | □ OTHERS |

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